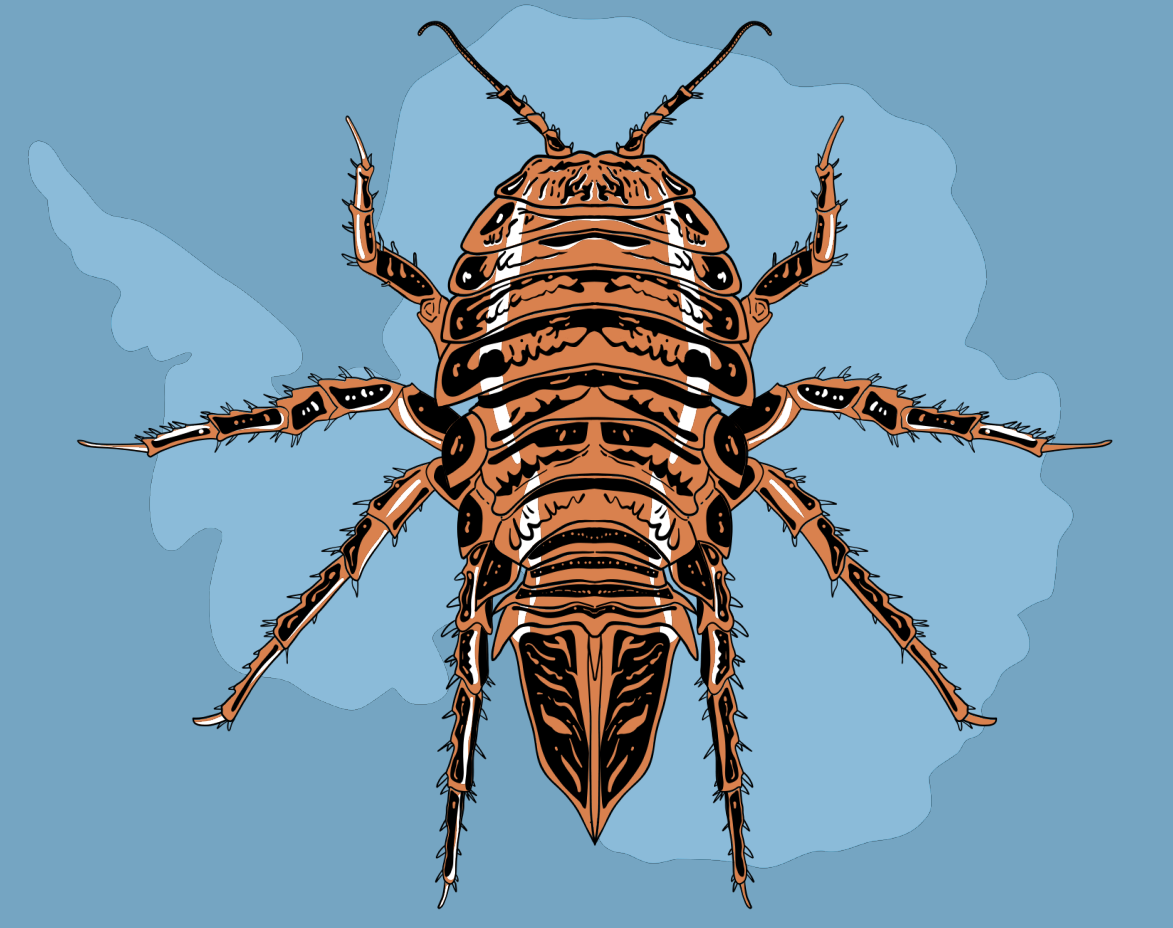


Unlocking Antarctic Species Trait Data Through Collaboration and Standardisation

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What are traits

Biological traits: morphology, physiology, behavior, ecology, life history

Taxonomic traits: hierarchical

Derived traits: human-defined, e.g. conservation status

Antarctic species are particularly trait-distinctive:

- Gigantism (sea spiders, isopods, sponges)
- Extreme slow growth and late maturity (toothfish, Antarctic krill)

Where to find traits

WoRMS / RAMS

- World Register of Marine Species - the global taxonomic standard for marine life
- RAMS = Register of Antarctic Marine Species - Antarctic subset, ~9,000 species
- Backed by thematic editors

FishBase / SeaLifeBase

- FishBase: traits for ~35,000 fish species; highly structured
- SeaLifeBase: invertebrates and other marine organisms
- Fixed team of editors

Why is trait information important

Fisheries

- Growth rate, longevity, age at maturity: stock assessment models
- Trophic level, diet: food web models
- Spatial distribution traits: defining sustainable fishing zones
- Reproductive strategy: estimating recovery time after exploitation
- The problem: Many bycatch and ecosystem species are lacking in trait data

Conservation

- Species richness alone can be misleading - 10 functionally unique species may matter more than 100 redundant ones
- Traits allow calculation of functional originality - how irreplaceable a species is
- Antarctic benthos has high functional originality due to its long evolutionary isolation
- Traits can contribute to species distribution models and food web reconstructions

The 2026 Antarctic species traits workshop

Fisheries reports, books, grey literature,...

Often paywalled, restricted access and unstandardized

- 11 participants registered 164 new species traits in WoRMS/RAMS from primary sources
- Five new traits were proposed for WoRMS/RAMS
- Discussions on a SCARFish initiative to make an inventory of Antarctic fish species traits

SO-fish Diversity

- An application to explore the diversity of fishes in the Southern Ocean
- Filter on fish length, IUCN status, feeding ecology,...
- Visualize diversity trends through time and space
- The app is still under development and maintained by the biodiversity.aq team

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